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Skin Mold Imaging Summer 2026 Technical Memorandum

Background

Transdermal Drug Delivery (TDD) is a non-invasive form of drug administration that features drug delivery through human skin and into the bloodstream. The minimally invasive nature of TDD makes it more comfortable and painless for patients, which often makes it a preferred treatment choice [1]. TDD is also desirable because its path of drug delivery does not pass through the liver and thus avoids first pass metabolism, which breaks down the drug before it reaches patients' bloodstreams [1].

However, the efficacy of TDD is limited by the human skin, which acts as a barrier between the environment and the rest of the human body. Human skin is made up of three layers: the outer layer called the epidermis, the blood vessel-filled middle layer called the dermis, and the lowest layer called the hypodermis [1]. The outermost layer of the epidermis is the SC, which consists of a brick-and-mortar framework. In this case, the lipids resemble the mortar and the dead skin cells called corneocytes resemble the bricks [2]. Molecules typically travel through both the corneocytes and the lipids in TDD.

Molecular transport across the SC can change with alterations to the SC's structure. Our lab explores how stretching the stratum corneum affects how substances transport through the skin. Since the SC contains non-uniform, microscopic furrows, stretching the skin should flatten out these furrows and impact transdermal transport. Stretching the SC may also widen the gaps between the corneocytes, which we hypothesize will increase the flux of molecules through the SC.

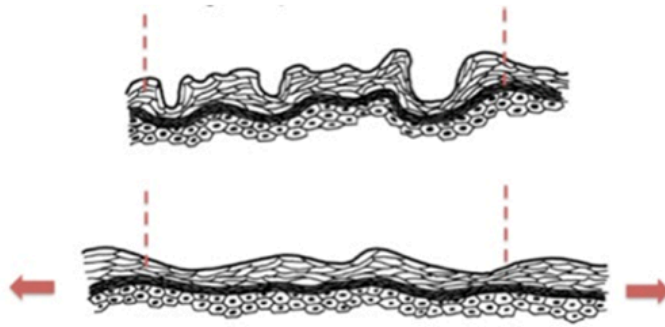


Figure 1: Stratum corneum furrows before (top) and after (bottom) uniaxial stretching

Our lab also hypothesizes that stretching the skin affects Transepidermal Water Loss (TEWL), which is the passive diffusion of water out of the stratum corneum and into the air. TEWL is often used to characterize skin permeability, with higher TEWL values indicating an increased flux of molecules in and out of the skin. TEWL values are also used to diagnose SC health, as molecules' ease of transport through the SC can demonstrate how effective the SC functions as a barrier.

TEWL is often measured with open-chamber and closed-chamber devices, which both have their own benefits and drawbacks. However, these TEWL meters do not take into account changes in skin surface area when reporting TEWL values as skin is stretched. So, to characterize these changes in surface area and normalize TEWL values, our lab is utilizing the skin molding and imaging process. Skin molds are created from ExaFlex, a silicone material often used for dental molds. When applied to the skin, the ExaFlex dries in around one to two minutes, resulting in a negative mold of the skin topography. These molds are then imaged with the Zeiss LSM510 Confocal Laser Scanning Microscope (CLSM) analyzed with image analysis softwares like Fiji (also known as ImageJ) to calculate surface area after image processing.

Initial Status

Prior to this summer, the skin imaging team has created procedures for taking, imaging, and analyzing skin molds. The previous teams typically mixed equal parts of the ExaFlex base and ExaFlex catalyst together on a watch glass to create a pink mixture to create the skin molds. This mixture is then placed onto a test subject's skin, where it dries in around 2-3 minutes. This mold is a negative skin mold, meaning that the ExaFlex fills in the furrows after being directly placed onto the skin. Negative molds contrast the positive molds that previous lab members used to create, which were molds of the negative skin molds. This additional step was performed to add dyes to the mold in order to see furrows with the CLSM without needing to recreate a new skin mold every time a dye was added. However, the Summer 2025 team discovered that the ExaFlex material of the skin molds fluoresces under the Argon laser without the need for added fluorescent dyes, so positive skin molds are no longer created.

Afterwards, the mold is superglued onto a small petri dish and filled with Milli-Q Water to prepare for imaging on the CLSM. The Summer 2025 cohort developed a [guide](#) for how to use the CLSM and how to process the resulting image Z-stacks in the Fiji software to obtain a surface area value from the BoneJ plugin.

This Summer 2025 team also obtained skin molds from six test participants – Subjects A through F. Two molds were created from each participant: one under stretched conditions sustained by our lab's uniaxial extensometer and one under neutral, unstretched conditions. Additionally, this cohort created SOPs for imaging these skin molds and using Fiji to perform image processing. However, these SOPs have only been used to obtain Z-stacks from each of the “Subject A” skin molds, so it is unclear whether

or not this procedure works for the other subjects' skin molds as well. None of the existing images from Subject A had been processed yet using Fiji, meaning that their corresponding surface areas have not yet been determined.

Work Done

This summer, lab members that worked on skin imaging first got more exposure to the entire skin molding process, as no previous skin imaging members were doing research in this summer cohort. The team started off by reading the procedures that have already been created and experimenting with software like Fiji.

After gaining a better understanding of the imaging process, the team met with two professors whose research focuses on computer imaging, Prof. Wloka and Prof. Mendelson, for advice on refining the imaging processing. The notes for these meetings can be found [here](#) under their corresponding tabs. After these meetings, the team experimented with different settings on Fiji which previous teams had not explored. These settings included binning to reduce image size, changing different binary methods, changing the radius for Gaussian blurs, and implementing Auto Focus. The results of these changes will be discussed further in the Results section.

During this time, our lab members also learned how to use the CLSM. Prof. Santana gave an introduction to how the CLSM worked, and the CLSM Summer 2025 guide provided our team with a reference for specific applications of the CLSM for our project. After familiarizing with the existing standard operating procedure for image processing, our team experimented with other settings to obtain the most ideal image for processing. By “ideal,” this means a clear image of the region of interest with a smooth gradient across the stack capturing the topography of the skin furrows. One change that worked was using the “Auto Exposure” button that was recommended by the Santana lab members (this was with their knowledge of laser microscopes in general; they have never used the CLSM before).

Additionally, a 3D-printed skin molding template was created to normalize the shape and size of our skin molds. The STL file and the SOLIDWORKS file for this template are located in the “Skin Mold Imaging” folder, which is located in the “Summer 2026” section of the lab Google Drive. With this new mold, a “control mold” was created, which was an ordinary skin mold placed onto a flat glass slide. Our intentions for doing this was to create a mold with a known surface area that would help us determine the optimal image processing settings and ranges for our other molds. Although the Fiji settings for the control mold may not transfer over completely because the surface is all on one plane, the general process should still be beneficial for improving our image processing procedure.

The skin molding template was also used to create skin molds of people over the age of forty, which began a new category for skin molds for our lab. Only three skin molds have been taken of people in this new group.

Results

First, the team experimented with the Fiji software, utilizing the binning option, which reduces the total number of pixels by combining neighboring pixels into one large pixel. This binning tool was explored because of the “OutOfMemoryError” that would appear after using BoneJ to calculate the surface area value for a Z-stack (Figure X). After binning the Z-stack and making the file size smaller, BoneJ was able to start outputting a surface area value. This memory error has not been noted in previous documentation, but it may just be because very few images were processed in the past.

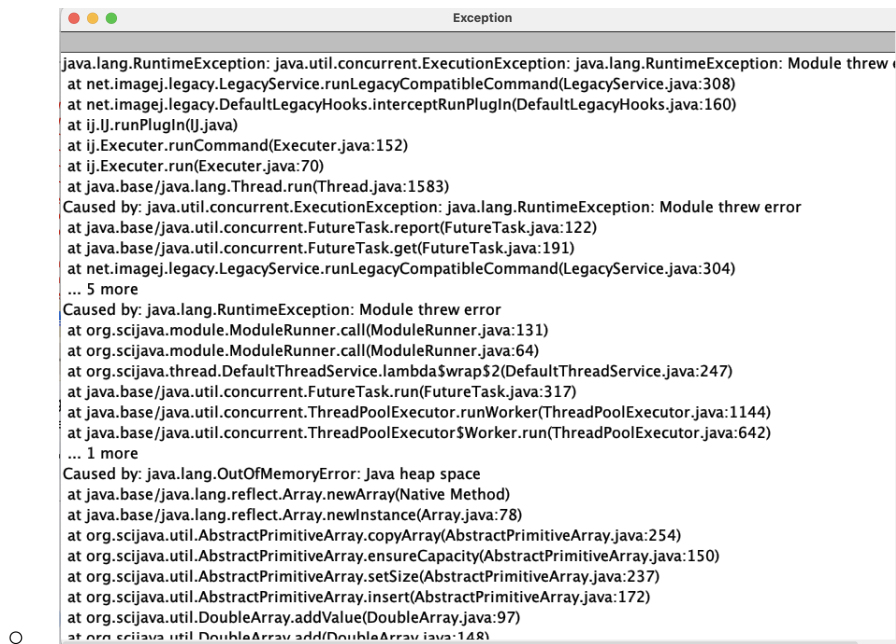


Figure X: Memory Error in Fiji After Calculating Surface Area with BoneJ

Our team members looked into various binarization methods. The Fiji image processing procedure created in Summer 2025 states that they used the “Huang” option for binarization, but no further details were included about the alternative options. So, our team experimented with the other

methods, and each of the seventeen methods seemed to have noticeably varying effects on the images. The optimal method for our project has yet to be discovered, including the binarization threshold range.

Our team also found that different sigma radii for the Gaussian blur step had an impact on the amount of noise in the images. From our experimentation, it was found that increasing the sigma radii decreased the amount of noise, as a wider range of pixels would apply to a blur (Figure B). However, increasing the sigma radii too much was found to even blur the desired pixels, so the blur should only be applied to the point where noise is mostly reduced. This test was only performed on the “SubjectAUnstretchedArea2” Z-stack that was collected in Summer 2025, where it was found that the ideal sigma radius for this specific image set was around 1.5 pixels.

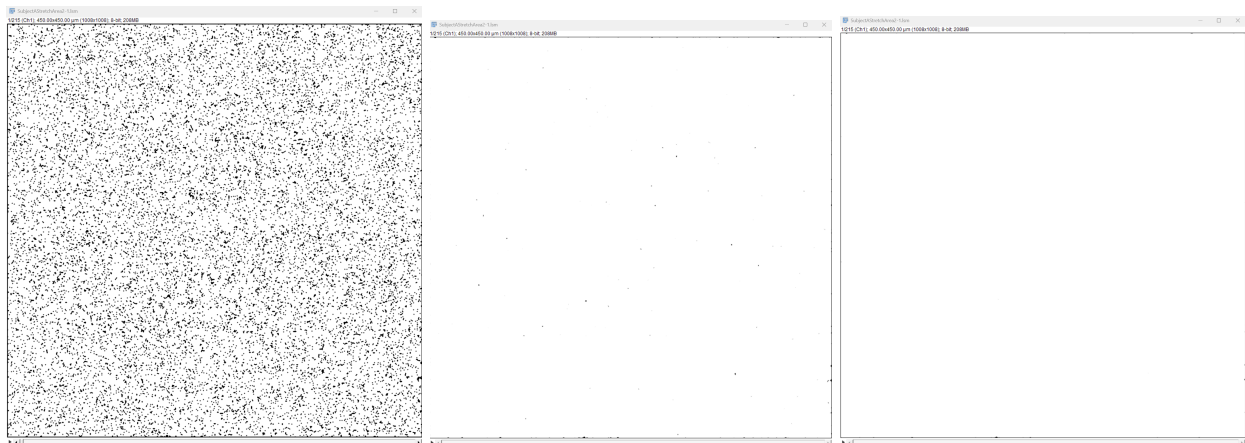


Figure B: (from left to right) Original image before adding a Gaussian blur, followed by two images with a sigma radius of 1 pixel and 2 pixels

Furthermore, turning on the “Auto Exposure” option on the CLSM allowed the microscope to display the skin furrows on the CLSM computer when the “Live” option was turned on. Without “Auto Exposure,” the live photos showed a purely dark red image even as the microscope stage is moved up and down (Figure A). Turning on “Auto Exposure” causes these photos to have a black background with red

dots dispersed across the image, with bright red splotches appearing at certain microscope stage positions (Figure A). Our lab members were originally hesitant to immediately conclude that these splotches are skin molds, as it was theorized that this may have been caused by the Super Glue that attaches the skin molds to the petri dish. However, Prof. Lape noted that the splotches would follow the same pattern throughout the Z-stack if they were super glue, leading us to be more confident in believing that these splotches are indeed skin furrows.

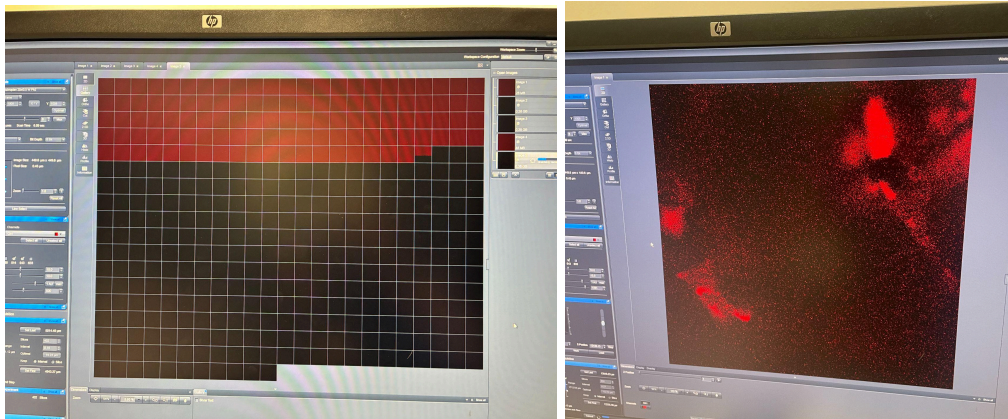


Figure A: CLSM Images Before Using Auto Exposure (left), and Images After (right)

Future Work

Future work would involve modifying the imaging and image analysis procedure to enhance the quality of the processed image to ultimately obtain 3D models and surface areas that are more precise. For the CLSM, the process of how “Auto Exposure” itself works in the microscope should also be looked into eventually, as our team has done minimal research on this setting.

For Fiji, more tests should be done on the binarization threshold and binarization methods. It is still unclear as to whether or not the range of values should stay the same for every Z-stack, so our lab plans to clarify this by imaging our control mold and seeing if these values can be used for other Z-stacks. After the threshold is researched, more research should be done on the binarization methods. From our work this summer, our lab members have found that changing the methods does indeed affect the image output noticeably, so combining this with more conclusive ranges for the binarization thresholds should be done to better reduce noise and obtain a more accurate surface area output.

Similar to the binarization thresholds, our lab should test if the ideal sigma radius for the Gaussian blur step stays constant across Z-stacks. In theory, the sigma radius should be different for skin molds due to variation in both the skin molds and the microscope measurement pathway (the “Set First” and “Set Last” values are pretty arbitrarily chosen). However, there is still a chance that this variation does not change our results significantly. Therefore, more Z-stacks should be processed to provide a concrete answer to this question.

In addition to adjusting specific settings and imaging more Z-stacks in general, changing the order of our image processing on Fiji should also be considered. This change was suggested by the professors our lab members spoke to this summer, but they have not been implemented yet due to time

constraints. So, it is worthwhile to go over the image processing procedure once more and determine whether some steps are better in a different order or even eliminated as a whole.

Finally, our team hopes to create more skin molds of people over the age of 40. Since this category of skin molds was created just this summer, very little data has been collected in this category. Therefore, more data should be collected to determine whether or not age plays a factor in the surface area calculations and TEWL measurements. Specifically, more stretched and unstretched skin molds from each participant over 40 years of age should be obtained and analyzed. These results would also help us achieve our goal of normalizing the TEWL measurements with our TEWL device in stretched and unstretched skin.

References

- [1] WY Jeong, M Kwon, HE Choi, KS Kim, “Recent advances in transdermal drug delivery systems: a review, ” *Biomaterials Research*, <https://pubmed.ncbi.nlm.nih.gov/34321111/>
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- [2] Wen, J., Koo, S. M., & Lape, N. (2018). “How sensitive are transdermal transport predictions by microscopic stratum corneum models to geometric and transport parameter input?” *Journal of Pharmaceutical Sciences*, 107(2), 612–623.
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